ADAM-Edu: 30-minute teacher practical

Explore LD, connect trait relationships to a breeding objective, then make a breeding decision using prediction accuracy, gain and diversity. Predict before looking at a result.

Before the workshop

You need a computer, a modern web browser and internet access. **No installation, R, programming or participant account is needed.** You can use any internet connection; you do not need to share the instructor's Wi-Fi.

1. Open the workshop page: https://lhmsai007.shinyapps.io/ADAM-Edu-current/workshop/index.html

2. Check that all three application buttons open. Save the manual and practical PDFs from that page. You do not need to read the whole manual before class.

3. Find your browser's Downloads folder. The applications exchange full population files through Download and Upload controls; separate tabs do not share populations automatically.

4. Work in pairs. Record or download each result before another run replaces it. Keep the application tab open while a simulation runs.

0-4 minutes: historical population and LD

Open Historical population. Inspect Population history, LD decay and Genetic diversity for Constant, Mild bottleneck and Severe bottleneck. Use the prepared LD figure on the workshop page if a new comparison is still running. Compare short and long genetic distances and expected heterozygosity. These prepared curves are one simulated replicate, not replicate means.

1. Does LD decrease with distance? Is one history highest in every bin?

2. Why can population numbers recover without restoring lost genetic variation?

Choose Mild bottleneck and download the full historical population. If needed, use the prepared historical population on the workshop page instead; it includes the LD results.

4-8 minutes: base population and breeding objective

Open Base population and upload that full history. Select the Mild bottleneck final scenario, choose B5 (negative genetic correlation) and click Generate teaching case. Inspect phenotype versus TBV. In BV and economic values, use true-BV ranking and 20% selected. Compare E2 (1,0) with E3 (2,1), without regenerating the animals. Save E3 with Use these EVs in Module 3 and download the full base for a later extended lesson.

1. Why can selection for Trait1 change Trait2? Which direction is favourable for each trait?

2. Changing economic values changes the objective and ranking. Does it change the animals' underlying breeding values?

The next task deliberately uses a separate prepared one-trait population. It is not a numerical continuation of B5. This gives both prediction methods the same controlled starting point.

Historical observation	Your evidence
Short- versus long-distance LD	
Bottleneck and diversity	

8-25 minutes: breeding programme and a reproductive decision

Run A: information for young candidates. Open Current population. Choose Workshop default: BLUP versus GBLUP (young candidates), then Load programme settings. Alternatively upload the prepared breeding population from the workshop page, then load the workshop default. Check: 6 years, 80 births/year, 4 age-1 sires, 80 regular dams aged 1-4, no MOET, 5 replicates and seed 2026. Keep Pedigree BLUP and GBLUP selected. Click Run breeding comparison once.

Young candidates have no own phenotype before age 2; their recorded relatives provide information. Both methods use the same available phenotypes, base, candidate numbers, selected-parent counts and reproductive settings. GBLUP additionally uses genomic relationships. Select young-candidate accuracy when inspecting Selection accuracy. It is the measured correlation between TBV and EBV before selection, not an assumed label.

Record both methods' realized accuracy, annual genetic gain, annual inbreeding rate and generation interval. Use the measured-comparison table in Workshop guide and inspect the curves and replicate uncertainty. Do not expect identical numbers in every run.

Run B: donor reproduction. Record A first. In Breeding schedule & selection change only the template: regular dams 40, MOET dams 4, MOET offspring share 50%, maximum offspring per MOET dam 10. Keep 80 total offspring, 4 sires and all other A settings. Apply the template to all years, check the table, then run again. This compares 40 conventional births plus 40 donor births with A's 80 conventional births. MOET here approximates extra offspring from selected donor dams; it does not model recipients or embryo physiology.

1. How much higher are realized accuracy and gain with genomic information in A? Why does limited own information matter?

2. How does B change gain and parental contributions? Does concentrating births in donor families change inbreeding?

3. Would you recommend B from gain alone? Compare diversity and generation interval. MOET does not automatically shorten the interval or lower inbreeding.

If runs are slow, use the prepared reference table for A and discuss the predicted trade-off in B. Label a prediction clearly; do not present it as an observed result. Avoid repeated clicks or larger settings during the timed session.

25-30 minutes: your teaching activity

Choose LD, heritability, correlated response, economic objectives, GxE, prediction or reproduction as your learning goal. State one input to change, one outcome to interpret and the prediction students must make first. Explain one model limitation and one breeding decision that requires more information than these simulations provide.

Results to record

Record an LD observation, a diversity observation, the E2/E3 correlated response, and A/B settings with method, seed and replicate count. For each breeding method record accuracy, gain, inbreeding rate and generation interval. Compare A and B using the same base, units and time period. Higher GBLUP accuracy does not guarantee lower inbreeding.